

PROJECT PROPOSAL

DEVELOPMENT OF A MULTIMODAL RETRIEVAL AUGMENTED GENERATION (RAG)-BASED INFORMATION RETRIEVAL SYSTEM FOR AGRICULTURAL EXTENSION SERVICES SUPPORTING NIGERIAN PIDGIN AND LOCAL DIALECTS

Introduction

Agricultural extension service is the backbone of food security in developing nations, serving as a bridge between scientific research and practical farming. In Nigeria, the effectiveness of these services is usually hampered by a high extension agent-to-farmer ratio, which currently stands at approximately 1:5000, which is far below the FAO recommendation of 1:800 (Olanrewaju et al., 2023).

The emergence of Large Language Models (LLMs) offers a proper solution. However, classic LLMs often lack localized knowledge and tend to suffer from “hallucinations”. Retrieval Augmented Generation (RAG) mitigates this by anchoring the model response in a verified database of agricultural documents (Lewis et al., 2020). Because Nigerian agriculture is diverse, a multimodal approach incorporating text and images of crop diseases is essential. Recent studies suggests that for AI to be truly effective in Africa, it must support socio-linguistic contexts, including Pidgin and major dialects like Hausa, Igbo and Yoruba (Adebayo et al., 2024).

Statement of the Problem

Despite the availability of agricultural innovations, most smallholder farmers in Nigeria remain underserved due to three barriers:

1. **Language Barrier:** Most digital tools are in English, whereas a significant portion of the farming population communicate in Nigerian Pidgin or local dialects.

2. Input Complexity: Traditional text-based systems are difficult for farmers with low literacy levels or those working in field conditions where capturing an image or sending a voice note is more practical.
3. Information access issues: Farmers struggle to access real-time, expert-verified advice on pest control and soil management, often relying on outdated practices.

Aim and Objectives

Aim

To develop a Multimodal RAG-based information retrieval system that provides localized, expert-verified agricultural extension support in Nigerian Pidgin and local dialects.

Objectives

- To design a multimodal vector database containing Nigerian-specific agricultural data.
- To implement a Retrieval Augmented Generation (RAG) pipeline that fetches context-aware information based on farmer queries.
- To integrate Natural Language Processing (NLP) modules for translating and processing Nigerian Pidgin and selected local dialects (Hausa, Igbo, Yoruba).
- To evaluate the system's accuracy, response time, and user satisfaction among smallholder farmers.

Significance of the Study

- Promotes Inclusivity: By supporting Pidgin and dialects, it brings cutting-edge AI to the "unconnected" rural farmer.
- Enhances Crop Yield: Providing instant, accurate diagnostics for pests and diseases can drastically reduce harvest losses.
- Reduces Costs: It complements the efforts of human extension agents, allowing them to focus on complex cases while the AI handles routine inquiries.

- Data Sovereignty: It contributes to the creation of localized datasets for Nigerian languages, which are often classified as "low-resource" in global AI research.

Methodology

The development will follow an Agile AI development lifecycle:

A. Data Acquisition & Pre-processing

- Source: Collating extension manuals from the Federal Ministry of Agriculture and Food Security (FMAFS) and International Institute of Tropical Agriculture (IITA).
- Localization: Translating key documents into Nigerian Pidgin and dialects using a mix of machine translation and human verification.
- Multimodal Integration: Collecting and labeling images of common Nigerian crop pests (e.g. Cassava Mosaic Virus,).

B. System Architecture

1. Embedding Layer: Using models like *CLIP* (for images) or *Whisper* (for audio/multilingual text) to convert queries into vectors.
2. Vector Database: Utilizing ChromaDB to store and retrieve high-dimensional agricultural data.
3. RAG Orchestration: Using a middleware layer to link the retrieved context with an LLM.
4. Translation Layer: Integrating a fine-tuned NLLB (No Language Left Behind) model to handle the dialect-to-English-to-dialect loop.

C. Implementation & Testing

The system will be deployed as a mobile-friendly web interface or mobile application, given its high penetration in Nigeria, with vigorous user-first testing involved.

Expected Result

- A Functional Prototype: A system capable of receiving a photo of a diseased leaf or a voice note in Pidgin and returning a diagnostic report in the same language.
- High Retrieval Accuracy: The RAG system is expected to achieve at least 80% accuracy in retrieving relevant scientific facts compared to a vanilla LLM.
- Reduced Latency: Optimizing the pipeline to ensure farmers receive responses within seconds, even on slower networks such as 3G networks.

User Empowerment: Evidence of increased confidence among farmers in making data-driven decisions regarding their farms.